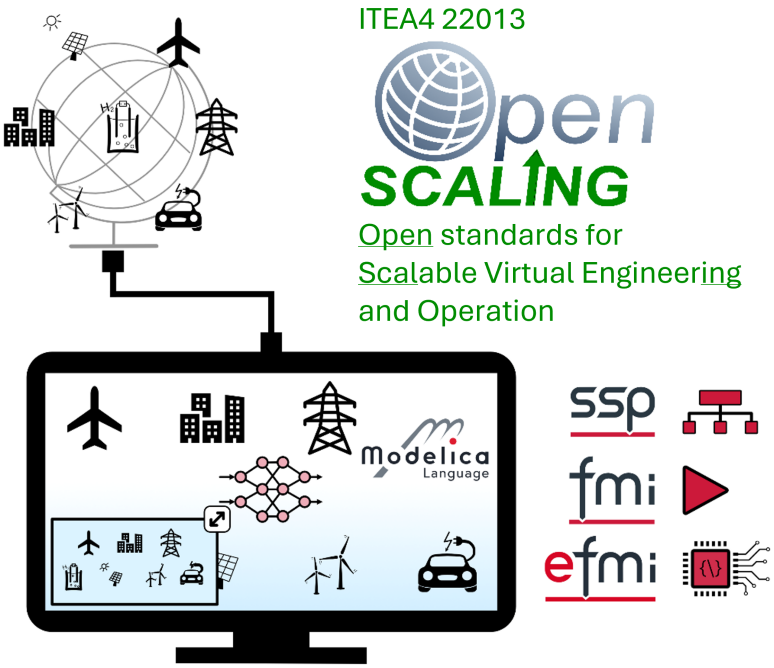
Contents

Deliverable D2.1: Uncertainty Quantification Draft . . . 2
1. Introduction . . . 5
1.1. Intent of this Document . . . 5
2. Examples . . . 5
2.1. DC-Motor . . . 5
2.1.1. Objective . . . 6
2.1.2. Model . . . 6
2.1.3. Uncertain Parameters . . . 6
2.1.4. Operational Domain . . . 7
2.1.5. Setup of Experiment . . . 8
2.1.6. Use of the XML-file for a forward UQ-experiment . . . 8
2.2. Industrial Mobile Valve . . . 9
2.2.1. Model . . . 9
2.2.2. Experiment Setup . . . 10
3. XML file format . . . 13
3.1. ExperimentSetup . . . 13
3.2. ParameterSet . . . 13
3.2.1. ParameterSetType . . . 14
3.2.2. UncertainParameter . . . 14
3.2.3. OperationalDomain . . . 16
3.3. ObservedVariables . . . 17
3.4. DesiredResults . . . 17
3.5. SimulationSetting . . . 18
3.6. SamplingMethod . . . 18
4. JSON file format . . . 19
5. Known Limitations of this Standard . . . 19
6. Contributors . . . 20
References . . . 20



Deliverable D2.1: Uncertainty Quantification Draft

Version 0.1 (September 30, 2025)

This document presents a draft proposal for defining *Uncertainty Quantification* in a standardized and tool-independent way. It is being developed as part of the OpenSCALING project under the ITEA4 program. The goal is to establish a new Modelica Association (MA) project which will produce a version 1.0 of this document as an MA standard and maintains it in the future.

AsciiDoc specific: By pressing "t", the table of contents displayed on the left side is toggled, so becomes hidden or is displayed again. Non-normative text is given in square brackets in italic font: *[Especially examples are defined in this style.]*

[Note: Although the document refers to major versions of the FMI, SSP, and Modelica Language standards, everything described in this document also applies to all subsequent minor versions. For further information on compatibility, see section Versioning and Layered Standards in the FMI 3.0 specification, as well as [Pre2013]] Copyright © 2025 The OpenSCALING consortium.

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This prototype draft is being worked on actively, and thus is subject to change without notice.

See also: LS-UQ Schema Extension Proposal (WP2/WP5 alignment: Assumptions, Domain Hierarchy, Activity Types)

Project Acronyms

Acronym	Description
DAE	Differential-Algebraic Equation
FMI	Functional Mock-up Interface (standard for model exchange and co-simulation, see https://fmi-standard.org/)
FMU	Functional Mock-up Unit (an instance of an FMI component, see https://fmi-standard.org/)
HVAC	Heating, Ventilation and Air Conditioning
LSS	Large-scale System
Modelica	Standard for modelling of cyber-physical systems, see https://modelica.org
ODE	Ordinary Differential Equation
SSP	System and Structure Parameterization

From Full Project Proposal

ID	Type	Description	Due Month	Access
D2.1	SW	Modelica, FMI, eFMI, SSP enhancements for UQ	M18	Public

1. Introduction

1.1. Intent of this Document

This document defines a tool-independent data model and related XML/JSON file formats for describing Uncertainty Quantification - Wikipedia (UQ) experiments. XML schema files are located in the schema directory. Script xml-to-json.js is provided to automatically convert an XML UQ definition into a compact JSON representation.

These file formats facilitate seamless exchange and execution of UQ workflows for models parameterized in a manner similar to the FMI standard, particularly targeting FMI, SSP and Modelica models. It is expected that tools will support importing and exporting these files to perform uncertainty quantification as defined in these files. Optionally, the XML file may also be stored in the `extra` directory of an FMI component in a *backwards compatible way* (= compliant tools can ignore definitions that they do not support).

The initial scope is intentionally narrow: Uncertainties are described at the parameter level and experiments are limited to *forward uncertainty propagation* with non-intrusive UQ-methods and *global sensitivity analysis* of potentially *nonlinear* dynamic models. Future versions will be expanded to include *validation against measurements* and *predictive-capability assessments*. The standard is designed for cross-layer reuse: For example, component suppliers can include UQ metadata together with their model (e.g., inside an FMU `extra` folder), while system integrators can compose these UQ artefacts in larger studies. Two examples - a DC motor and an industrial mobile valve - illustrate the format and its intended uses.

Non-normative text in section 2 and section 3 is given in square brackets in italic font: *[Especially examples are defined in this style.]*

[Note: Although the document refers to major versions of the FMI, SSP, and Modelica Language standards, everything described in this document also applies to all subsequent minor versions. For further information on compatibility, see section Versioning and Layered Standards in the FMI 3.0 specification, as well as [Pre2013]]

2. Examples

2.1. DC-Motor

The following example demonstrates how to perform a UQ experiment on an electric motor provided as Modelica model `DCMotorUQExample.ScenarioDCMotor` in the file examples/dc_motor/DCMotorUQExample.mo.

The section is structured as follows:

- The objective of the UQ experiment is formulated in the dedicated Objective section,
- The simulation model used for the UQ experiment is documented the dedicated Model section,
- The UQ experiment is defined with information documented in dedicated sections:

- The Uncertain Parameters,
- The Operational Domain,
- The Setup of Experiment.

The information related to the UQ experiment is hosted as .xml-file or .json-file (= automatically generated from the XML file), refer to the following link for the DC-motor example:

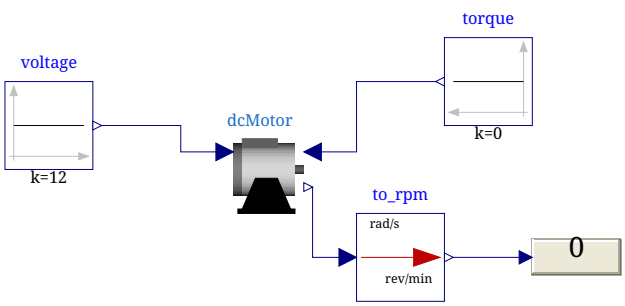
- XML file: dc_motor_setup.xml,
- JSON file: dc_motor_setup.json.

2.1.1. Objective

The objective is to determine the uncertainty in the motor speed (= `dcMotor.omega`), given the uncertainties of some of the motor parameters.

2.1.2. Model

The top-level diagram of this model is shown in the next figure:



The component `dcMotor` has the following parameter menu:

Parameters

R	0.1	Ω	resistance
L	1e-4	H	inductance
c_m	0.01	N·m/A	electrical (back-EMF) motor constant
c_g	0.01	N·m/A	mechanical (torque) motor constant
J	5e-5	kg·m²	inertia of the rotor
d	1e-4	N·m·s/rad	friction

The motor is initially at rest and accelerates due to the input voltage `voltage.k = 12V` until it reaches a steady state speed.

2.1.3. Uncertain Parameters

For a UQ experiment, each model parameter is defined in the XML file as *superset* of element <uq:ParameterSet> of the SSP standard:

- The definition of a deterministic parameter remains unchanged, refer e.g., to the parameter `dcMotor.d` documented below.
- The definition of an uncertain parameter contains following additional information, refer e.g.,

to the uncertain parameters `dcMotor.R` and `dcMotor.L` documented below:

- The stochastic distribution associated to the uncertain parameter e.g., *Normal*, *Uniform*,
- The derivation of the defined stochastic distribution e.g., *Estimated* or *Computed*,
- The data source for the characterization of the distribution e.g., *From project XYZ*.

With this definition, it is possible to sample the uncertain parameters randomly according to their respective distributions distribution and to the selected sampling method and run a simulation with the simulation model based on this sampling.

```
Unresolved directive in ls-uq.adoc - include::examples/dc_motor_parameters1.xml[]
```

Please note, that:

- The namespace `uq:` is used for all new elements introduced by this document.
- The namespaces `ssv:` and `ssc:` are from the SSP standard.

2.1.4. Operational Domain

The UQ experiment is performed in an operational domain (n-dimensional space with n axes) where the model shall be valid. Please note, that the definition of this domain is optional). For the DC-motor example documented below, these axes are the excitation voltage `voltage.k` and the external disturbance torque `torque.k`.

Please note, that the parameters `voltage.k` and `torque.k` can be seen as uncertain parameters that cannot be characterized by stochastic distributions due to insufficient knowledge. This type of uncertainty is known as epistemic uncertainty (see, for example, [Dan2020]).

The operational domain is delimited by boundaries, defined via the element `Boundary` and is fully defined by the convex hull of given boundary points, defined by `Point coordinates`. For the DC-motor example documented below, the operational domain is exemplarily delimited by four points (1V, 0Nm), (1V, 5Nm), (10V, 0Nm), (10V, 5Nm).

Within this domain, it is possible to perform a UQ experiment with a given UQ-method in different subdomains via the element `ExperimentPoints`, each subdomain being defined via the element `PointSet`. For the DC-motor example documented below, two subdomains have been defined:

- `PS1` delimited by three points `P1`, `P2`, `P3`,
- `PS2` delimited by three points `P4`, `P5`, `P6`.

Please note, that each point of a subdomain has a unique identification `id` used to identify the results for this point.

```
Unresolved directive in ls-uq.adoc - include::examples/dc_motor_parameters2.xml[]
```

2.1.5. Setup of Experiment

It is possible to fully setup the UQ-experiment with following information:

- The observed variables (also called Quantities of Interest). Each variable is listed as element `ObservedVariable` in the element `ObservedVariables`. Optionally, it is possible to specify that the result at one time point shall be approximated by a *Normal* distribution via the attribute `distributionApproximation`. For the DC-motor example below, the observed variable is `dcMotor.omega` and has to be approximated as `Normal` distribution.
- The results to be computed for all observed variables and the name of the result datafile (element `DesiredResults`). Please note, that it is possible to provide the name of a directory if results are stored in several files. For the DC-motor example below, the trajectory of the observed variable is of interest. Following results have to be saved in `dc_motor.mat`:
  - The mean trajectory, the related standard deviation respectively the percentile levels (`0`, `5%`, `50%`, `95%`, `100%`),
  - The histogram, the Probability Density Function - PDF - and the Cumulative Density Function - CDF, respectively,
  - The 2nd order and the total Sobol indices, respectively for global sensitivity analysis purposes.
- The integration method to be used for the simulations (element `SimulationMethod`). Optionally, tool-specific setup data can be provided with the `Annotation` element using for example key/value pairs defined with the standardized element `KeyValue`. For the DC-motor example below, the simulations have to be performed with the `DASSL` solver configured with following settings:
  - Stop time: `0.5` s,
  - Interval: `0.01` s,
  - Tolerance: `1e-4`.
- The UQ-method used to sample the uncertain parameters (element `SamplingMethod`). For the DC-motor example below, the UQ-experiment has to be performed with the `Latin Hypercube Sampling` UQ-method configured with following settings:
  - Number of samples: `1000`,
  - Random number generator: `MT19937-64`,
  - Random number seed: `10`.

```
Unresolved directive in ls-uq.adoc - include::examples/dc_motor_experiment.xml[]
```

2.1.6. Use of the XML-file for a forward UQ-experiment

Please note, that to make the forward UQ-experiment runnable for the DC-motor example, python packages have to be installed from this Download requirements.txt list.

The UQ-experiment consists mainly of following activities:

- Sample the uncertain parameters with a given UQ-method: Please refer to the related example Download uncertain_parameter_sampling.ipynb file to inquire how to generate the samples.
- Perform the simulation with the values of the sampled uncertain parameters and post-process the results: Please refer to the related example Download Forward_UQ_DC_motor_example.ipynb file to inquire how to run the simulation and display.

Please note, that following .py-scripts are used to read automatically the content of the XML-file and parse it to the .ipnb-files:

- Download uq_xml_param_file.py: Returns information about the uncertain parameters and their distributions.
- Download uq_xml_forward_config.py: Returns information about the UQ-method and its settings.

2.2. Industrial Mobile Valve

This example describes a forward UQ experiment carried out on a work section of a Parker Hannifin mobile hydraulic valve [Par2024].

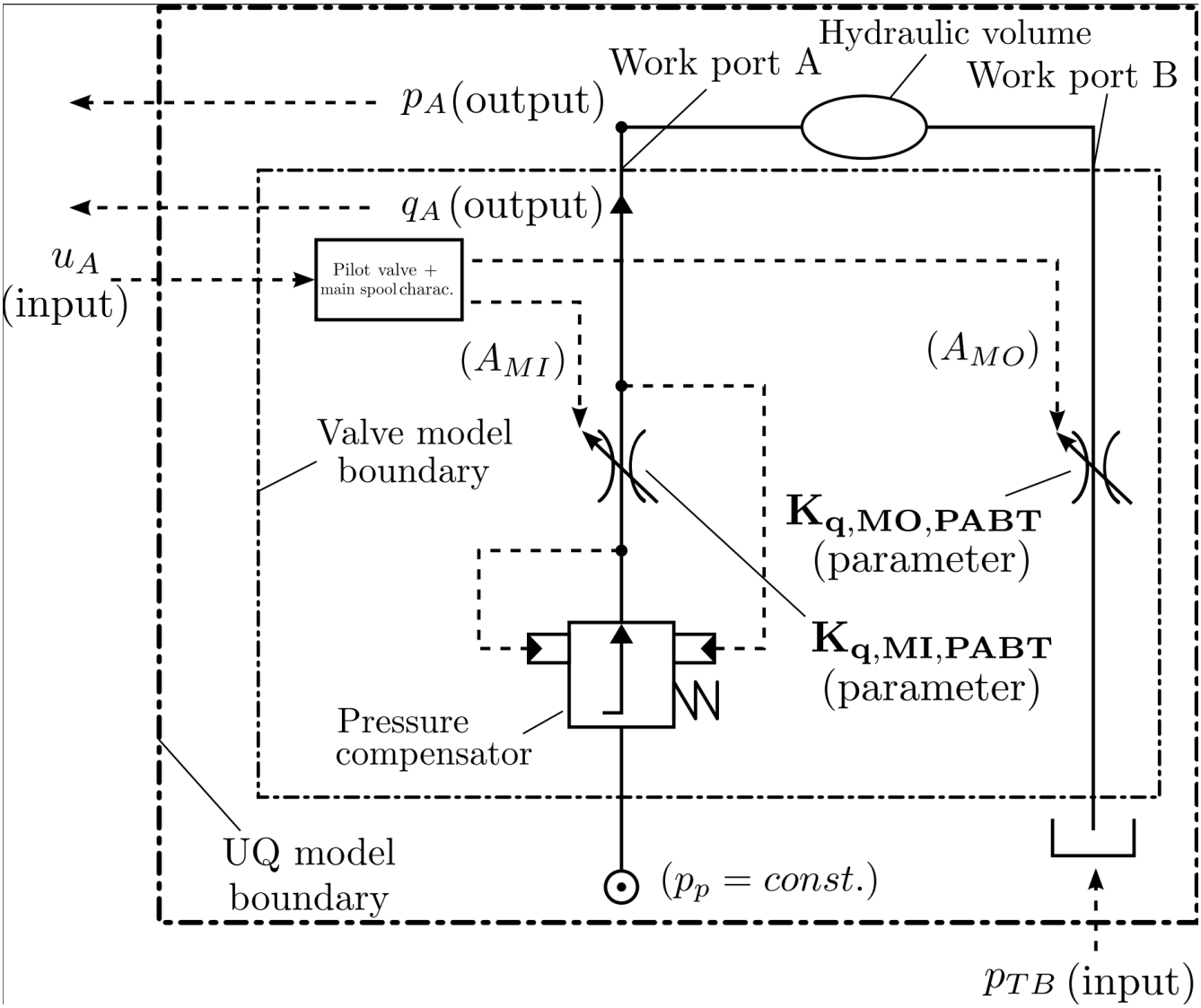
2.2.1. Model

This use case is based on a single work section of a proportional, load-sensing, pre-compensated mobile directional control valve from the Parker L90LS series [Par2024]. The valve's core function is to deliver a hydraulic flow that is proportional to an input electrical current, regardless of variations in pump and load pressures.

In a typical simulation scenario, the valve's work ports are connected to a closed hydraulic volume and supplied with constant pump pressure. The flow and counterpressure responses are evaluated for different input currents.

Manufacturing tolerances introduce uncertainty in parameters such as the discharge coefficient. Therefore, a Uncertainty Quantification (UQ) study is performed to assess the effect of these parameter uncertainties on valve behavior. This example demonstrates how UQ metadata can be packaged with the valve model using the proposed standard format.

The next figure shows the schematics of the model of the example mobile valve, which is packaged as an SSP component in file examples/mobile_valve/mobile_valve.ssp:



2.2.2. Experiment Setup

We are interested in how uncertainty in the valve gains, Kq,MI,PABT and Kq,MI,PABT propagate to uncertainty in the supplied flow (qA) and counter pressure (pA) outputs. The operational domain is assumed to be 0-900 uA control current (uA) and 0-20 bar supply pressure (pTB). This is an operational domain in two axes, i.e. a box, but more complex convex hulls are possible with multiple axes and boundary coordinates. The uncertainties of the valve gains are normally distributed and also the supplied flow and counter pressure are expected to be normally distributed. To establish the distribution of qA and pA, 10 different points in the operational domain are defined. For each of these, 1000 simulations will be done according to the defined "Simulation method" and with varying valve gains. Based on this, the two distributions can be established.

The uncertain parameters of the mobile valve and the experiment setup are defined in the XML file examples/mobile_valve/mobile_valve_setup.xml as follows:

```
<?xml version="1.0" encoding="UTF-8"?>

<uq:ExperimentSetup
  name="ScenarioMobileValve_ForwardUQ"
```

```
xmlns:uq="http://openscaling.org/UQ1/UncertaintyQuantification"
xmlns:ssv="http://ssp-standard.org/SSP1/SystemStructureParameterValues"
xmlns:ssc="http://ssp-standard.org/SSP1/SystemStructureCommon">

<uq:Model
  name="ScenarioMobileValve"
  file="ParkerSimPack.ssp"
  modelResourceMetaData="mobile_valve.srmd" />

<uq:ExperimentSet>
  <uq:ForwardUncertaintyQuantification>
    <uq:ParameterSet>
      <uq:Parameters>
        <uq:UncertainParameter name="KqMI_PABT_gain" source="Estimated"
          sourceInfo="Parker Hannifin's Hydraulic Valve Systems">
          <uq:Normal mu="1" sigma="0.1" minimum="0.8" maximum="1.2" unit="-" />
        </uq:UncertainParameter>
        <uq:UncertainParameter name="KqMO_PABT_gain" source="Estimated"
          sourceInfo="Parker Hannifin's Hydraulic Valve Systems">
          <uq:Normal mu="1" sigma="0.1" minimum="0.8" maximum="1.2" unit="-" />
        </uq:UncertainParameter>
      </uq:Parameters>

      <uq:OperationalDomain>
        <uq:Axes>
          <uq:Axis name="uA" unit="mA" />
          <uq:Axis name="pTB" unit="bar" />
        </uq:Axes>

        <uq:Boundary>
          <uq:ConvexHull>
            <uq:Point coordinates="0 0" />
            <uq:Point coordinates="0 20" />
            <uq:Point coordinates="900 20" />
            <uq:Point coordinates="900 0" />
          </uq:ConvexHull>
        </uq:Boundary>

        <uq:ExperimentPoints>
          <uq:PointSet id="PS1">
            <uq:Points>
              <uq:Point id="P1" coordinates="345 1" />
              <uq:Point id="P2" coordinates="355 1" />
              <uq:Point id="P3" coordinates="375 1" />
              <uq:Point id="P4" coordinates="395 1" />
              <uq:Point id="P5" coordinates="415 1" />
              <uq:Point id="P6" coordinates="440 1" />
              <uq:Point id="P7" coordinates="465 1" />
              <uq:Point id="P8" coordinates="485 1" />
              <uq:Point id="P9" coordinates="505 1" />
              <uq:Point id="P10" coordinates="525 1" />
```

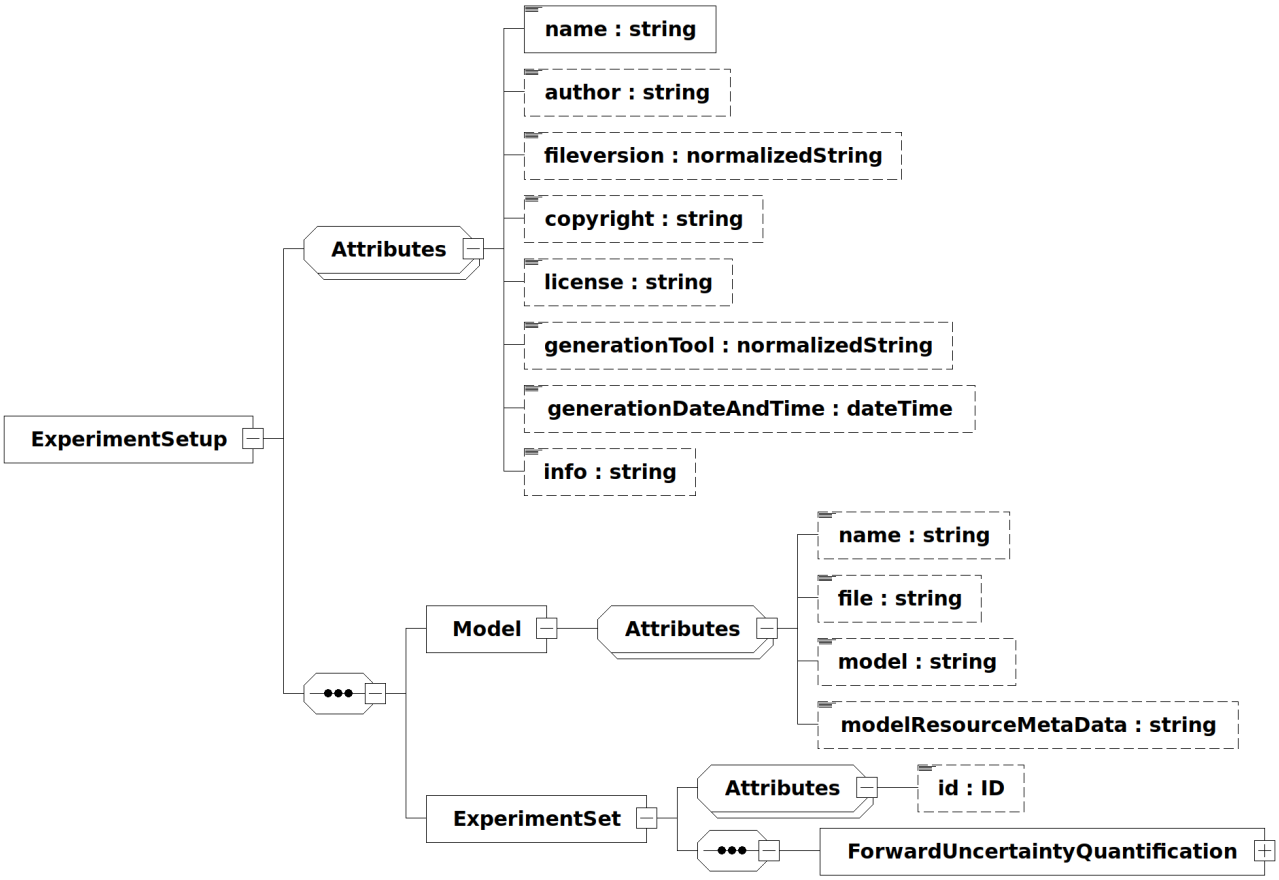
```
              </uq:Points>
            </uq:PointSet>

          </uq:ExperimentPoints>
        </uq:OperationalDomain>

        <uq:Units>
          <ssc:Unit name="-">
            <ssc:BaseUnit factor="1" offset="0" />
          </ssc:Unit>
          <ssc:Unit name="bar">
            <ssc:BaseUnit kg="1" m="-1" s="-2" factor="1e5" offset="0.0" />
          </ssc:Unit>
          <ssc:Unit name="mA">
            <ssc:BaseUnit A="1" factor="1e-3" offset="0.0" />
          </ssc:Unit>
          <ssc:Unit name="L/min">
            <ssc:BaseUnit m="3" s="-1" factor="6e-4" />
          </ssc:Unit>
        </uq:Units>
      </uq:ParameterSet>

      <uq:ObservedVariables>
        <uq:ObservedVariable name="qA" distributionApproximation="Normal" />
        <uq:ObservedVariable name="pA" distributionApproximation="Normal" />
      </uq:ObservedVariables>

      <uq:DesiredResults
        data="mobile_valve.csv"
        summary="mobile_valve_result_summary.xml"
        scope="FinalTime"
        histogram="true">
      </uq:DesiredResults>

      <!-- TODO: Viktor -->
      <uq:SimulationMethod stopTime="0.5" interval="0.001" tolerance="1e-4"
solver="FixedStepMaster" />

      <uq:SamplingMethod>
        <uq:PseudoRandom numberOfSamples="1000" />
      </uq:SamplingMethod>
    </uq:ForwardUncertaintyQuantification>
  </uq:ExperimentSet>
</uq:ExperimentSetup>
```

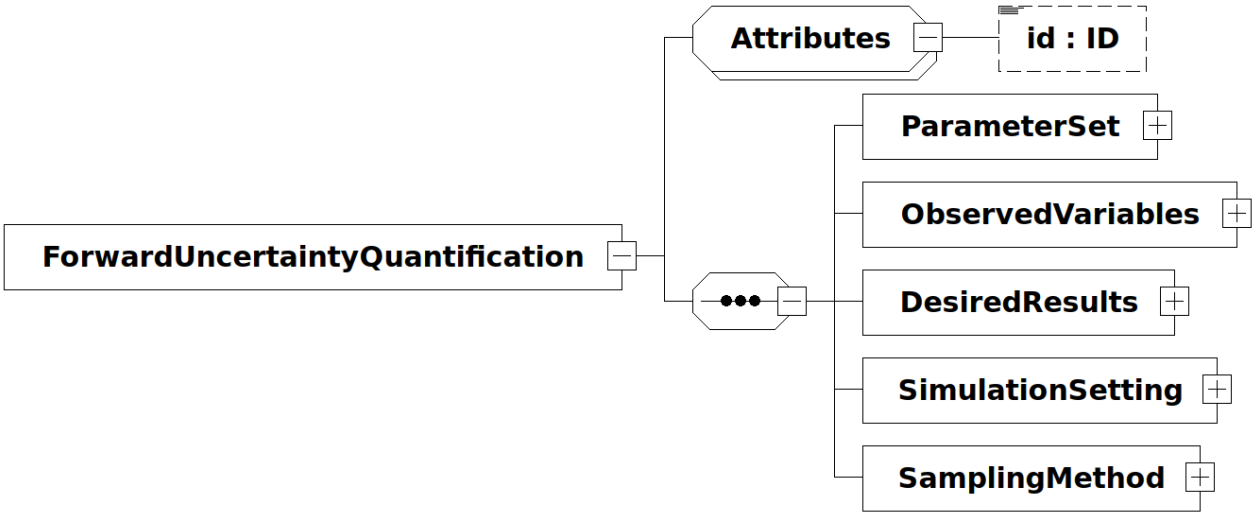
More details of this example and the results of the forward UQ experiment for the requested experiment points are provided in [Baj2025].

3. XML file format

This section defines the uncertainty quantification with XML schemas.

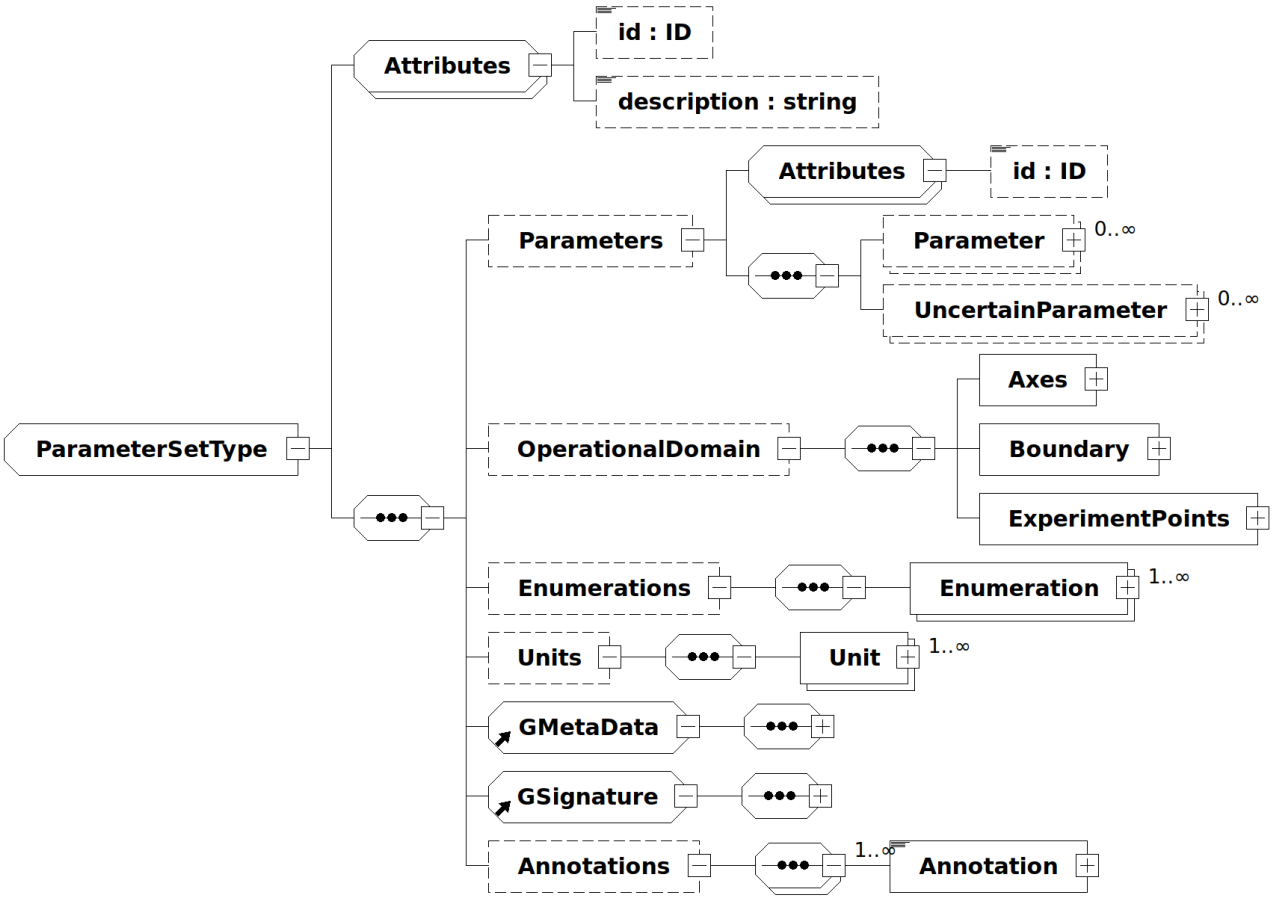
3.1. ExperimentSetup





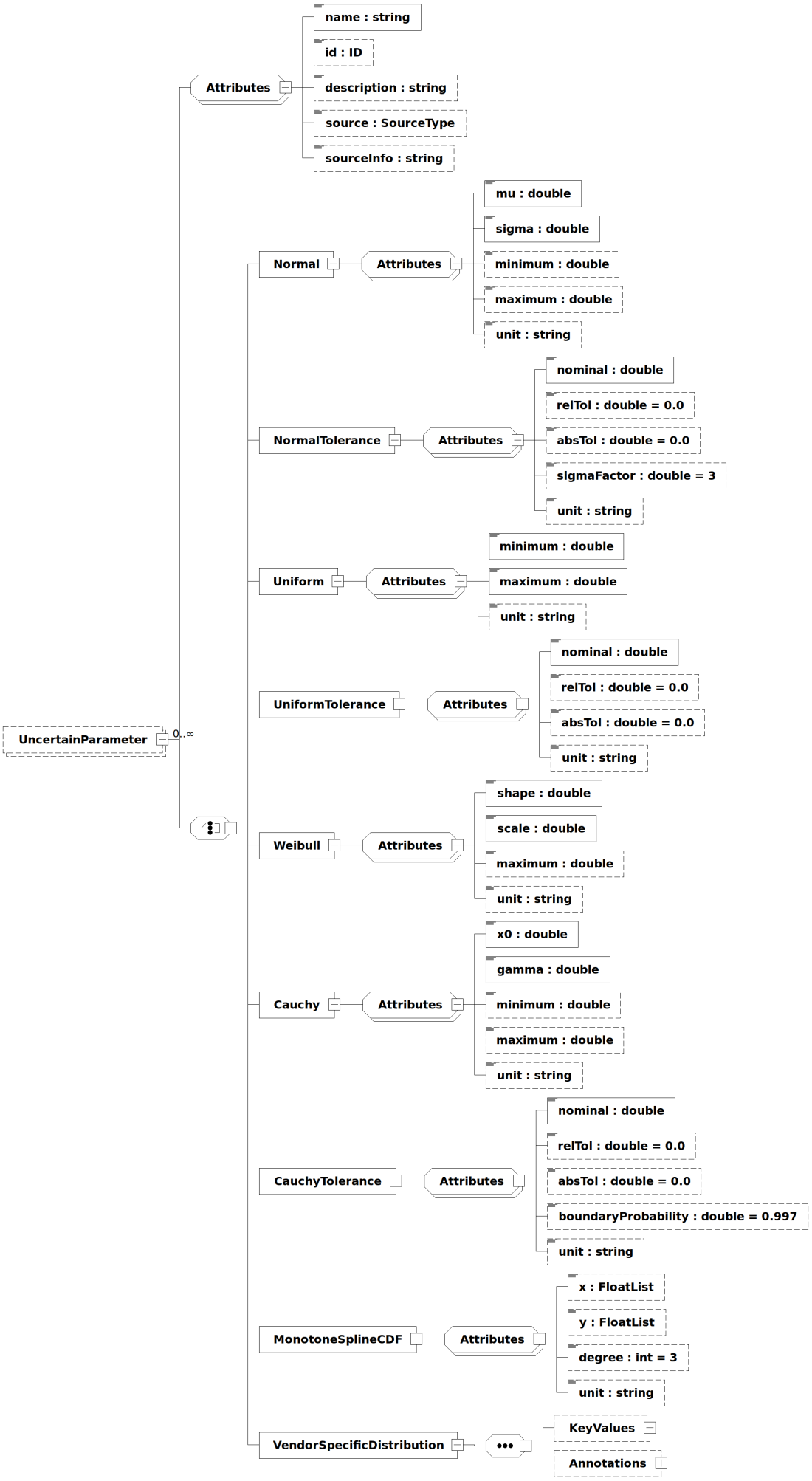
3.2. ParameterSet

3.2.1. ParameterSetType



All elements are from the SSP standard, with exception of UncertainParameter and OperationalDomain. The definitions of these elements are given below.

3.2.2. UncertainParameter


UncertainParameter
0..∞
Attributes
name : string
id : ID
description : string
source : SourceType
sourceInfo : string
Normal
Attributes
mu : double
sigma : double
minimum : double
maximum : double
unit : string
NormalTolerance
Attributes
nominal : double
relTol : double = 0.0
absTol : double = 0.0
sigmaFactor : double = 3
unit : string
Uniform
Attributes
minimum : double
maximum : double
unit : string
UniformTolerance
Attributes
nominal : double
relTol : double = 0.0
absTol : double = 0.0
unit : string
Weibull
Attributes
shape : double
scale : double
maximum : double
unit : string
Cauchy
Attributes
x0 : double
gamma : double
minimum : double
maximum : double
unit : string
CauchyTolerance
Attributes
nominal : double
relTol : double = 0.0
absTol : double = 0.0
boundaryProbability : double = 0.997
unit : string
MonotoneSplineCDF
Attributes
x : FloatList
y : FloatList
degree : int = 3
unit : string
VendorSpecificDistribution
KeyValues
Annotations

Uncertain scalar floating variables are described by continuous, stochastic distributions. These can be, for example, uncertain parameters as input for an experiment or distributions that variables reach in steady-state as result of an experiment.

> **Missing (needs to be added):**
> **array floating variables, scalar integer variables**

Three different types of distributions can be defined (for details, see image and schema/UncertainParameterType.xsd)

1. **Standard distributions**, such as the `Normal` distribution, represent widely known distributions that are defined with a few parameters. Via the *optional* attributes `minimum` and `maximum` it is possible to define lower and upper bounds on every distribution, transforming them into *truncated* distributions. The given parameters are always with respect to the underlying original, non-truncated, distribution. *[A tool can redraw a random number from which a quantile is computed, if the quantile is not in the minimum..maximum limits, or it can transform the underlying distribution to its truncated form with a simple formula, see e.g. Wikipedia, Julia function Truncated.]*
2. The distribution `MonotoneSplineCDF` defines a *Cumulative Distribution Function (CDF)* by a monotone spline that is defined by node values (x_i, y_i) with y_i the Cumulative Distribution values and a spline `degree`. It is required that all values x_i and y_i are strictly monotone increasing. If, for example, either measurements or Monte Carlo simulation CDFs are available by **tabulated values**, a monotone spline of the desired degree can be constructed for example with the open source Python function `PchipInterpolator` from `scipy.interpolate`. The corresponding *Quantile* can be determined by constructing a monotone spline using the interchanged node values (y_i, x_i) and the *Probability Density Function (PDF)* can be constructed by differentiating the montone spline of the CDF.
3. The distribution element `VendorSpecific` defines a distribution that is not (yet) standardized in this document but is available in the target tool. It allows a **target tool specific** definition.

Attributes of Distributions:

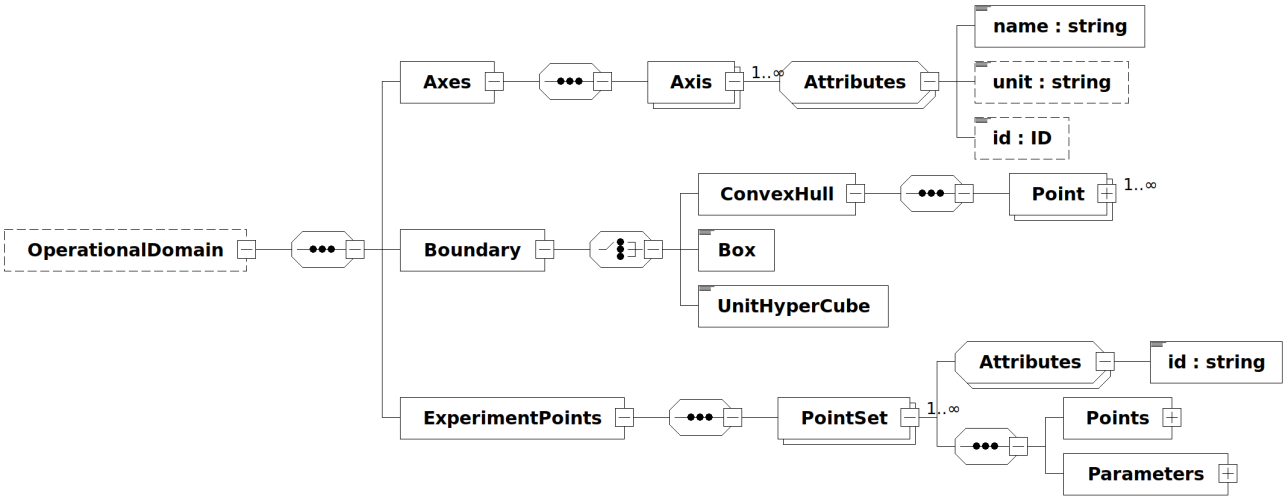
Show attributes in a table.

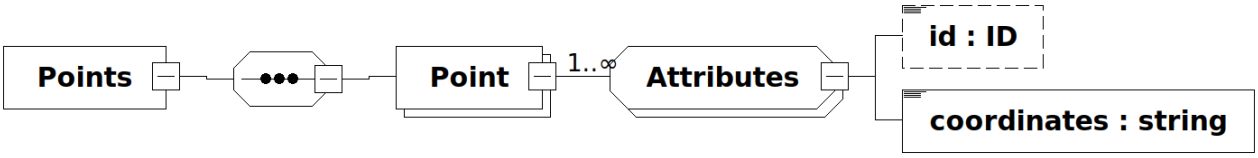
Test of Formulas:

PDF: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$

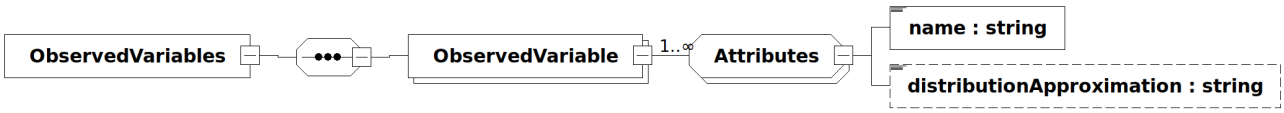
Quantile: $Q(p) = \sqrt{2}\,\sigma\,\mathrm{erf}^{-1}(2p-1) + \mu$

3.2.3. OperationalDomain

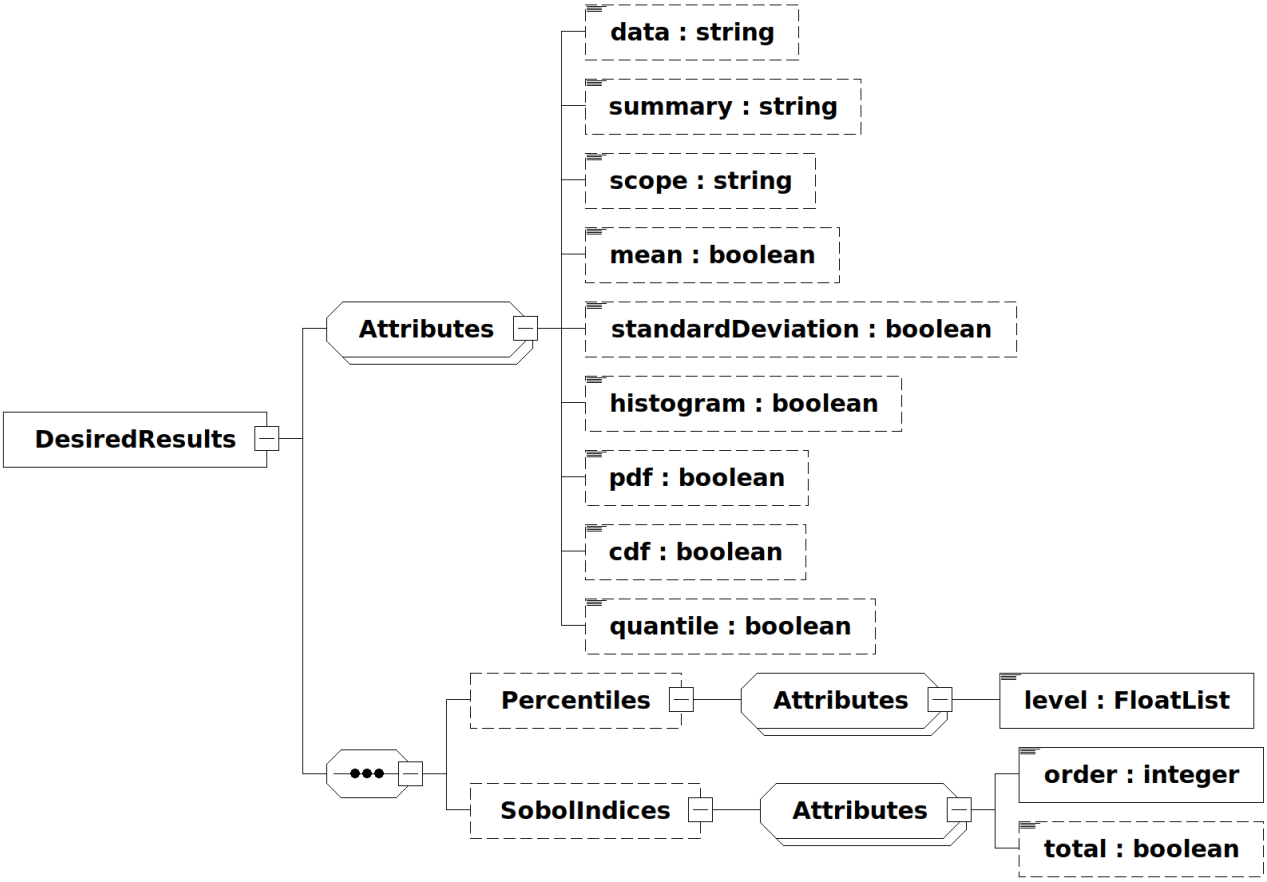




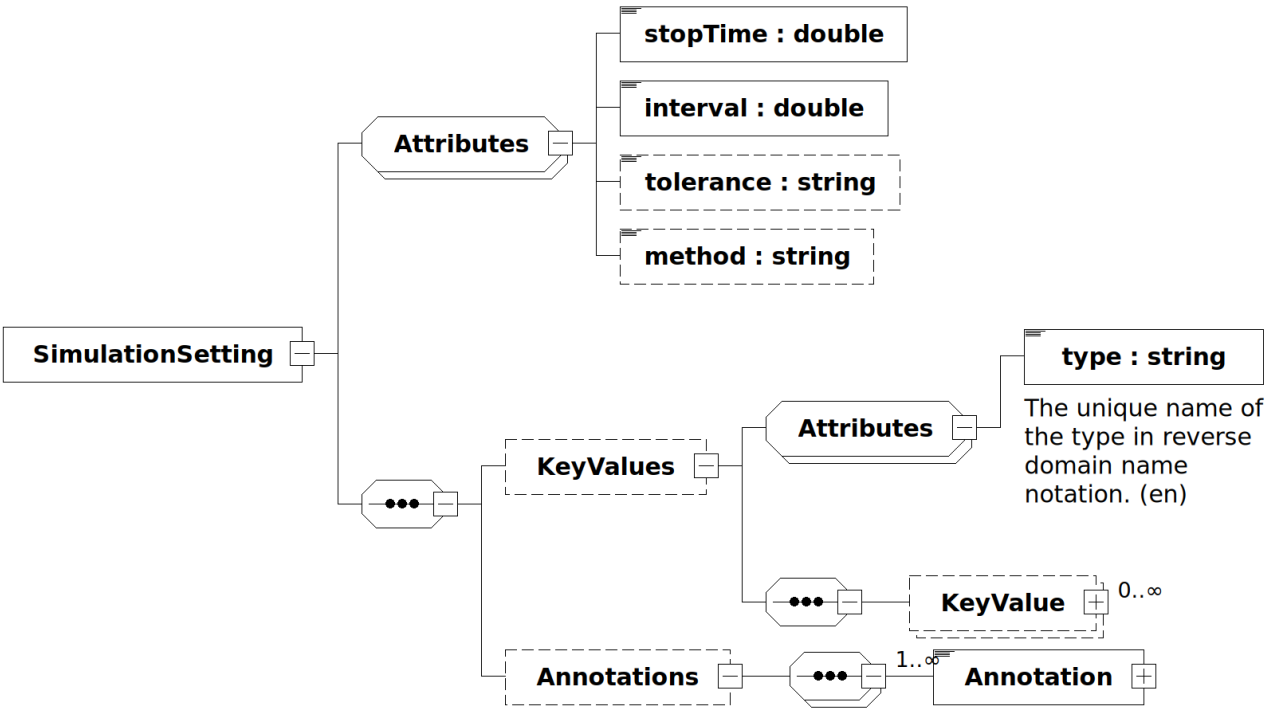
3.3. ObservedVariables



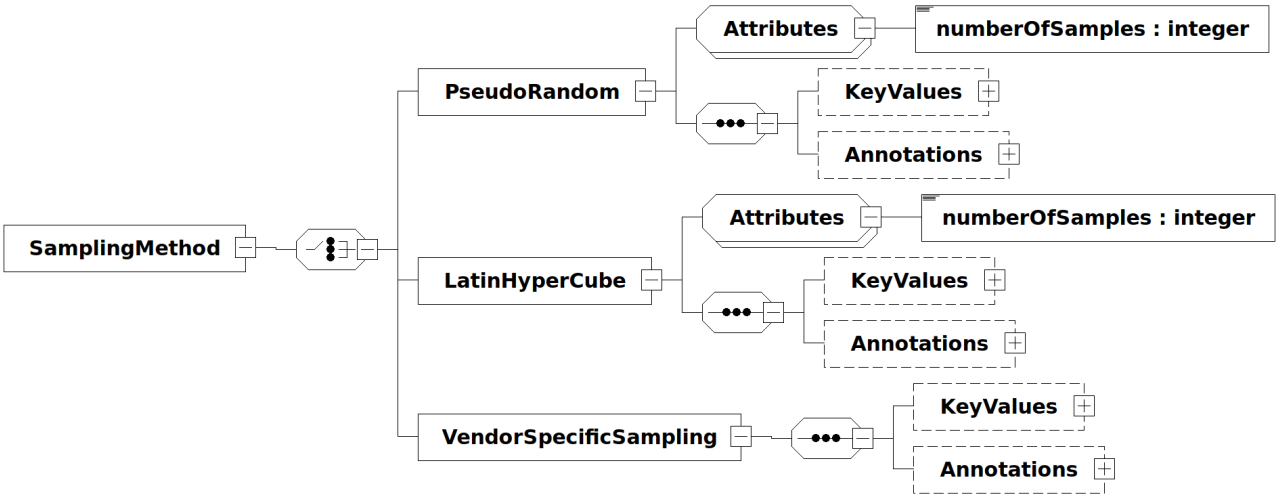
3.4. DesiredResults



3.5. SimulationSetting



3.6. SamplingMethod



4. JSON file format

A JSON file can be automatically generated from an XML file with the provided script scripts/xml-to-json.js using the command:

```
node  xml-to-json.js  file.xml  file.json
```

This script uses xml-js for the conversion with the xml-js options:

- compact: true,
- ignoreDeclaration: true,
- ignoreComment: true

Additional Javascript code performs further transformations

- Expands any attributes objects into their parent objects (removing the attributes layer) and placing these elements before any element on the same level
- XML string values are mapped to JSON values, such as
  - "0.2" is changed to 0.2 (number)
  - "true" is changed to true (boolean)
  - "0.2 0.3 4 5" is changed to [0.2, 0.3, 4, 5] (number vector)

5. Known Limitations of this Standard

To be reviewed

- Computational Scalability: Monte Carlo sampling methods scale poorly with model complexity and parameter count.
- Limited UQ Methods: Standard covers forward propagation only. Missing: inverse UQ, global sensitivity analysis, etc.

- Tool Implementation Dependencies: Success requires vendor adoption of tool-specific annotations. No enforcement mechanism for consistent implementation across platforms.
- Schema Extensibility: Current data model may not accommodate future UQ methodologies or domain-specific requirements.

6. Contributors

- Amin Bajand, Linköping University, Sweden
- L. Viktor Larsson, Parker Hannifin Manufacturing, Sweden
- Lena Buffoni, Linköping University, Sweden
- Elmir Nahodovic, Dassault Systèmes, Sweden
- Robert Hällqvist, Saab Aeronautics, Sweden
- Oliver Lenord, Robert Bosch GmbH, Germany
- Hans Olsson, Dassault Systèmes, Sweden
- Martin Otter, German Aerospace Center (DLR-FK), Germany
- Antoine Vandamme, Robert Bosch GmbH, Germany
- Adrian Pop, Linköping University, Sweden

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